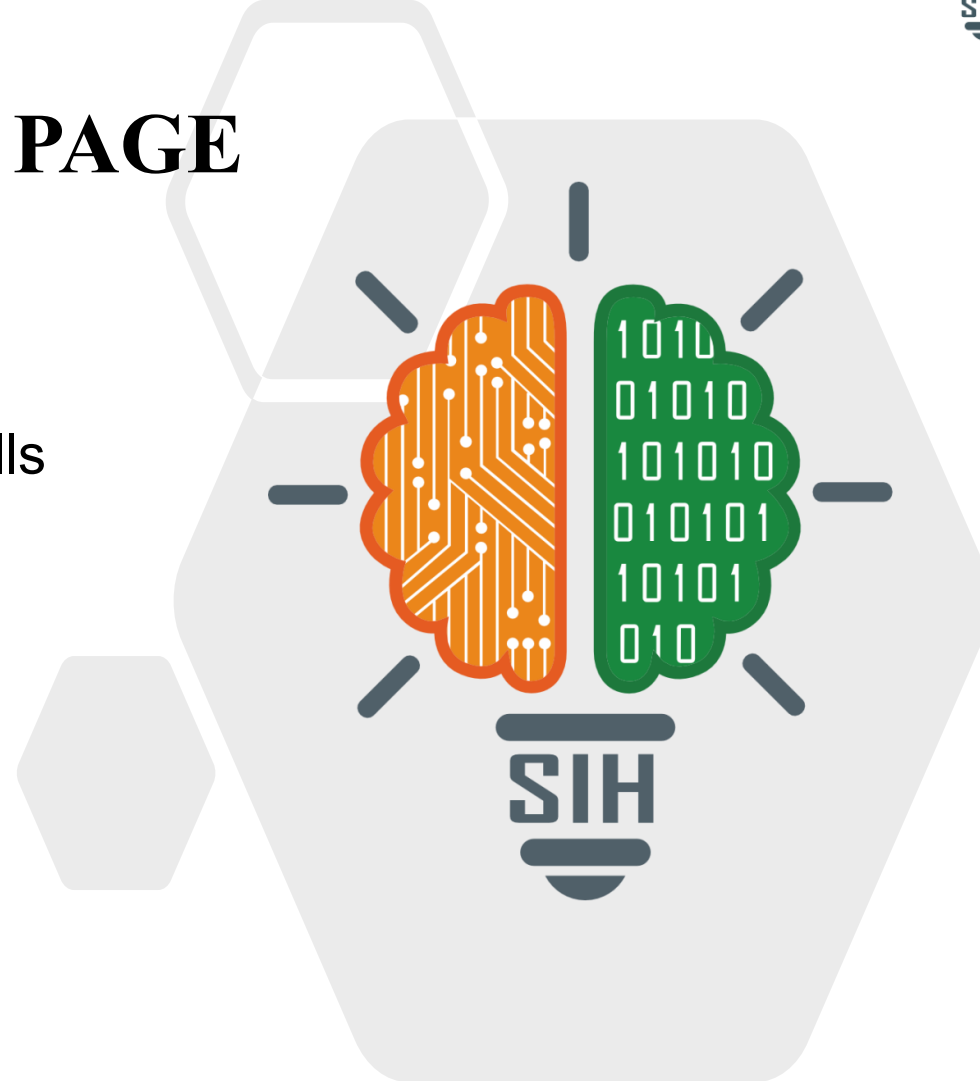


TITLE PAGE

- **Problem Statement ID** – SIH26121
- **Problem Statement Title** – Nearby Wells Intelligence System (NWIS)
- **Theme** – **as listed on the SIH portal**
- **PS Category** – Software
- **Team ID** – **Team ID**
- **Team Name** – **Team name**



NWIS · Nearby Wells Intelligence System

Drilling decision support beside eRTMAC · Oil India Limited

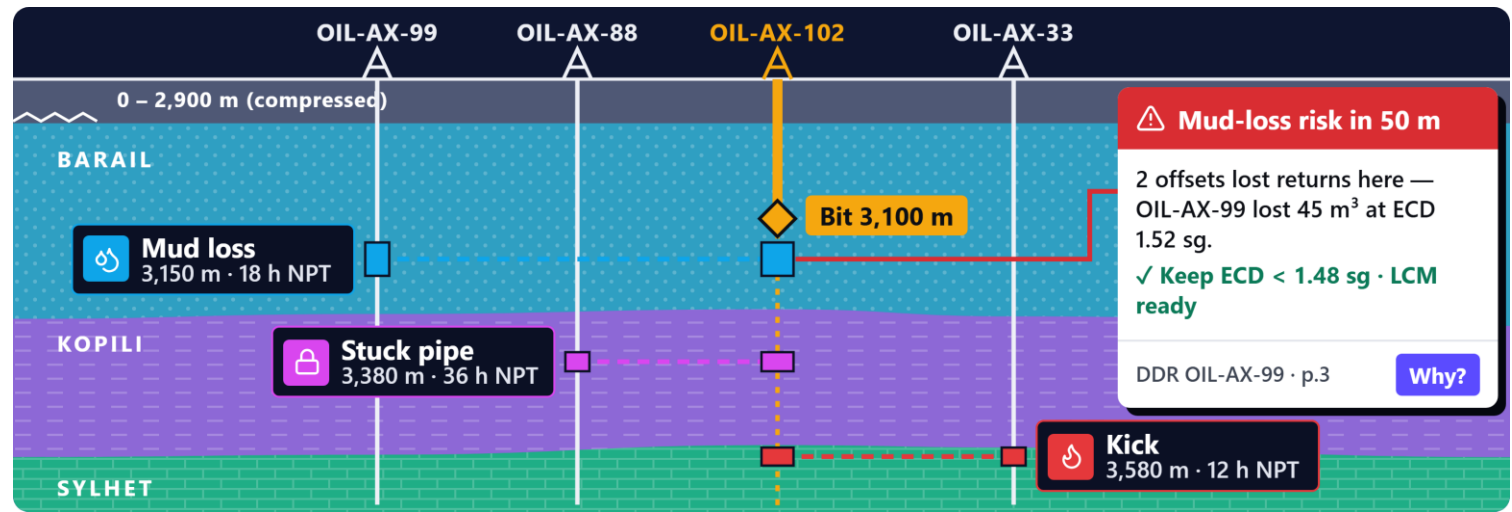
Team
Name

NWIS — Nearby Wells Intelligence System

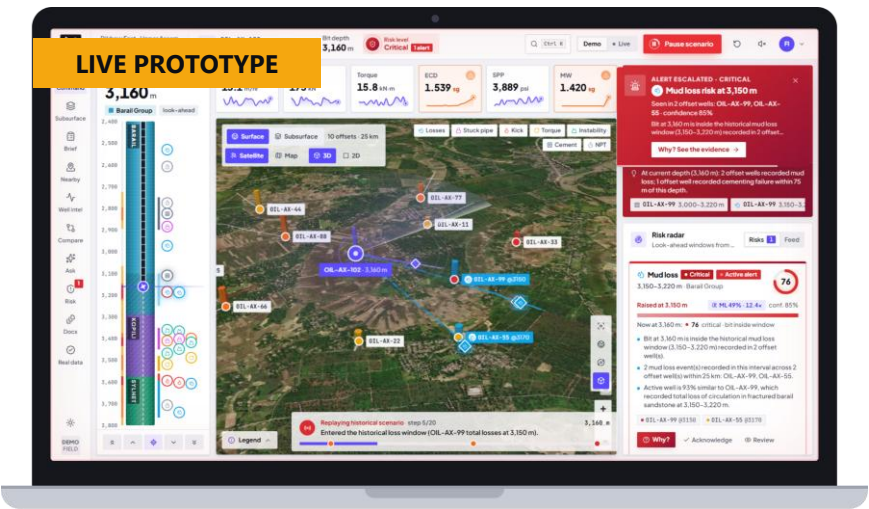


❖ Proposed Solution (Describe your Idea/Solution/Prototype)

1 Detailed explanation of the proposed solution



Links the active well's **bit depth** to what **offset wells** hit at that depth — and warns before the bit gets there.



2 How it addresses the problem

✓ All 20 official requirements PASS

Nearby wells on a 3D map, any radius

Offset history in seconds — every line cited

Correlates depth, formation, casing & mud

Warns before the bit reaches a risky depth

Prototype
<add link>

Video
<add link>

GitHub
[sih26121](#)

3 Innovation and uniqueness of the solution

Depth-linked
bit depth drives map, 3D, risk, search

Evidence-first
every answer cites report, page, depth

Explainable + ML
6-factor score + ML check (AUC 0.86)

Look-ahead brief
next 300 m for the shift handover, ₹ NPT

Real-data proof
1,970 real well histories, 94 % precision

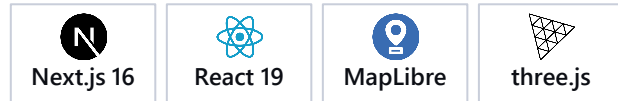
Team
Name

TECHNICAL APPROACH

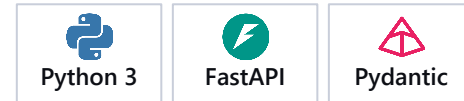


1 Technologies to be used

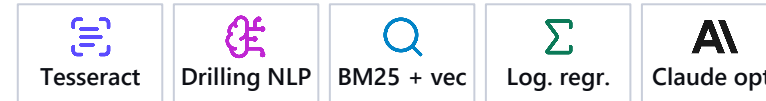
Frontend



Backend



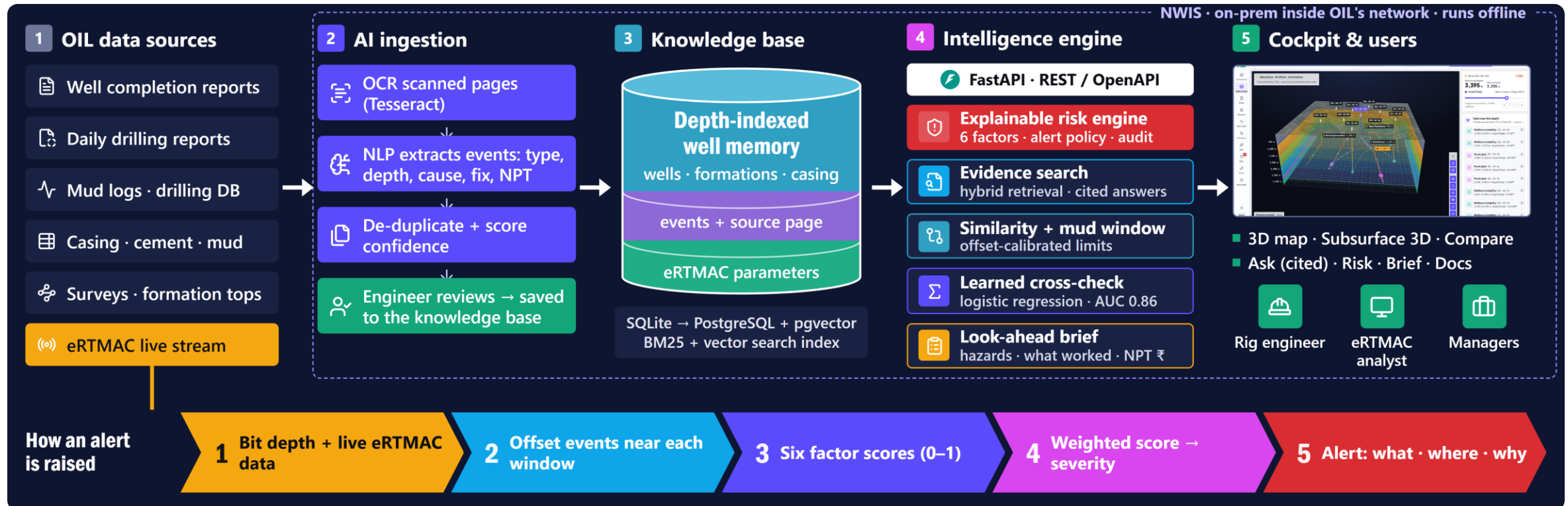
AI · NLP · OCR



Data & integration



2 Methodology and process for implementation



Team
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FEASIBILITY AND VIABILITY



1 Analysis of the feasibility of the idea

20/20

official SIH26121
requirements pass an API
check

94 %

precision on real, held-out
well histories

0.86

AUC of the learned risk
cross-check

< 30 ms

for a cited answer or a full
risk evaluation

- ✓ **Technical** — open source, one server, no GPU
- ✓ **Operational** — beside eRTMAC, no workflow change
- ✓ **Economic** — no licence cost; saves NPT days

2 Potential challenges and risks



Confidential well data

High



**Scanned, messy old
reports**

High



Alert fatigue on the rig

Medium



Trust in AI advice

Medium



Demo-tuned weights

Low

3 Strategies for overcoming these challenges



On-prem in OIL's network, **runs offline**; the LLM is optional and **off by default**; SSO roles.



OCR with per-page confidence; **an engineer approves** every extracted event.



Weak windows need **live confirmation**; one alert per window — it **escalates, never repeats**.



Every alert shows **Why?** — factors, cited evidence, confidence. **The engineer decides.**



Refit on OIL's NPT history, then a **shadow pilot** before alerts go live.

Viability — from prototype to OIL rigs in 12 months

NOW

Prototype ready

20/20 requirements · real-data proof ·
demo film

MONTH 0-3

Ingest one field

OIL DDR / WCR archive, SME review, refit
weights

MONTH 3-6

Shadow pilot at eRTMAC

2 rigs · alerts logged, not shown ·
measure lead

MONTH 6-12

Roll out

live WITSML feed · SSO · all Assam rigs

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IMPACT AND BENEFITS



1 Potential impact on the target audience



Rig engineer & driller

Sees what offsets hit at this depth — 50 m before the bit gets there.



eRTMAC monitoring team

Live data linked to offset history; cited evidence in seconds.



Planners & DWOP

Look-ahead brief: hazards, mud window and NPT in ₹ per section.



New engineers & managers

The field's memory stays searchable; every alert is audited.

Engineers don't lack data. They lack the link between **this depth, right now** and **what happened here before**.

Why it matters

≈ 20 %

of well-construction time is lost as non-productive time (NPT) [2]

up to 40 %

of NPT comes from kicks, losses and wellbore instability — exactly what NWIS warns about [3]

≈ ₹21 cr / yr

if NWIS saves one NPT day on each of the ~70 wells OIL spuds a year [4]*

* 1 day × 70 wells × ₹30 lakh/day assumed spread rate — an illustration, not a forecast.

2 Benefits of the solution



Economic

Fewer repeated losses, stuck pipe and kicks → less NPT, faster planning.



Safety

Earlier kick and overpressure awareness supports well control.



Environmental

Fewer loss events → less mud, LCM and cement lost; less rig diesel.



Social

Expert knowledge stays in India's energy PSU; faster onboarding.

1

Research — where today's approaches fall short

✓ yes

○ partly

✗ no

	Manual offset review	Monitoring alone	Generic chatbot	NWIS
Tied to the live bit depth	✗	✓	✗	✓
Reads DDR / WCR text, even scans	✓	✗	○	✓
Cites report page + depth	○	✗	✗	✓
Warns before the risky depth	✗	○	✗	✓

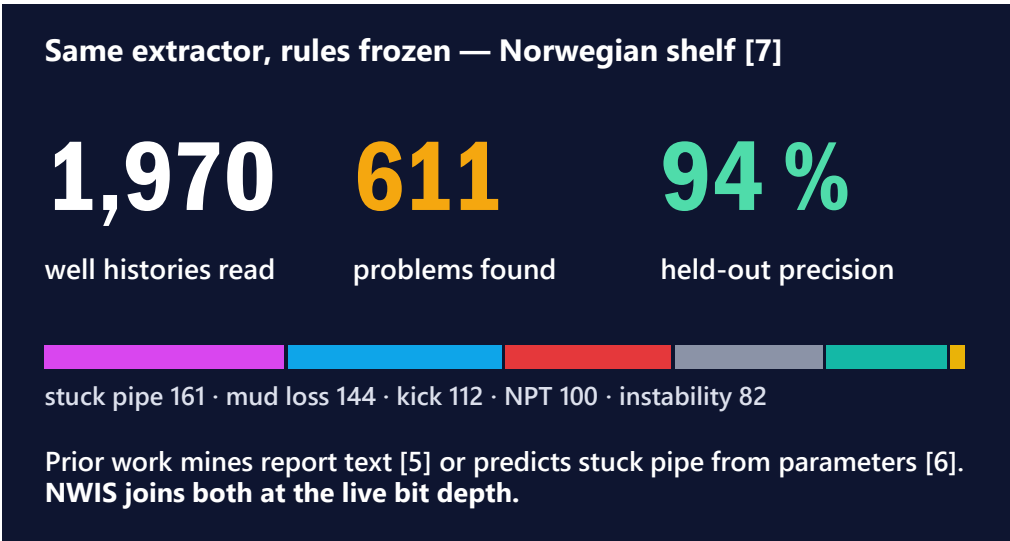
3

Details / links of the reference and research work

- [1] SIH26121 Nearby Wells Intelligence System — Oil India Limited · [SIH 2026 portal](#)
- [2] NPT ≈ 20 % of well-construction time — [Drilling Contractor, 2018](#)
- [3] Kicks, losses and instability: up to 40 % of NPT — [Offshore Magazine, 2011](#)
- [4] OIL: 70 wells spudded in FY25, 21 rigs — [Chairman's speech, 66th AGM, 2025](#)

2

Validated on real public well records



- [5] Hoffimann et al., Pattern mining in drilling reports with deep NLP — [arXiv, 2017](#)
- [6] Montes, Ashok, van Oort, Review of stuck-pipe prediction — [SPE Journal, 2025](#)
- [7] Norwegian Offshore Directorate well data, NLOD 2.0 — [Sodir FactPages](#)
- [8] WITSML real-time drilling data standard (eRTMAC adapter) — [Energistics](#)